Lecture 9. Random Number Generation

Congruential generators, inverse CDF, Box–Muller, acceptance–rejection, multivariate sampling

Contents

9.1 Uniform random numbers **2**

9.2 The inverse CDF method **2**

9.3 Normal random numbers **3**

9.4 Generating from definitions **3**

9.5 Acceptance–rejection sampling **3**

9.6 Multivariate distributions **4**

9.1 Uniform random numbers

All random number generation starts from i.i.d. Unif$(0,1)$ variates. Computers produce none: they produce deterministic **pseudo-random** sequences that only look random.

Multiplicative congruential generators. Take a large prime modulus m and a multiplier $a \in \{2, \dots, m-1\}$, and iterate $x_{k+1} = ax_k \bmod m$, outputting $u_k = x_k/m$.

Proposition 9.1

The map $f(x) = ax \bmod m$ is a bijection of $\{1, \dots, m-1\}$. The sequence is periodic with period $n = \min\{k \geq 1 : a^k \equiv 1 \bmod m\}$, which divides $m-1$.

Proof. $\gcd(a, m) = 1$ because m is prime, so ax is not a multiple of m for $1 \leq x \leq m-1$: f maps the set into itself. If $ax \equiv ay$, then $m \mid a(x-y)$, hence $m \mid x-y$ and $x = y$: f is injective, hence bijective on a finite set. Since $x_k = a^k x_0 \bmod m$, the sequence returns to x_0 exactly when $a^k \equiv 1$. By Fermat's little theorem $a^{m-1} \equiv 1 \bmod m$, so the period is at most $m-1$ (and divides it). ■

x_0 is the **seed**. If a is a **primitive root** modulo m (period $m-1$), the sequence runs through all of $\{1, \dots, m-1\}$. A classical choice is $m = 2^{31} - 1$ (a Mersenne prime) and $a = 7^5 = 16807$, with period $2^{31} - 2$ (used in early MATLAB).

A good generator needs a long period and output that passes statistical tests of uniformity and independence. Congruential generators have a structural flaw: consecutive p-tuples $(u_k, \dots, u_{k+p-1})$ lie on a small number of parallel hyperplanes in $[0,1]^p$ (Marsaglia's "random numbers fall mainly in the planes"; the notorious RANDU generator puts all triples on 15 planes). Shuffling helps. R's default, the **Mersenne Twister** (Matsumoto–Nishimura, 1998), has period $2^{19937} - 1$ and is equidistributed in up to 623 dimensions.

9.2 The inverse CDF method

For a cdf F define the generalized inverse $F^{-1}(u) = \inf\{x : F(x) \geq u\}$, $0 < u < 1$.

Theorem 9.2

Let $X \sim F$ and $U \sim \text{Unif}(0,1)$.

(a) $F^{-1}(U)$ has cdf F.

(b) If F is continuous, $F(X) \sim \text{Unif}(0,1)$.

(c) In general, $P\{F(X) \leq y\} \leq y$ for $y \in [0,1]$.

Proof. (a) We show $\{F^{-1}(u) \leq t\} = \{u \leq F(t)\}$. The set $\{x : F(x) \geq u\}$ is an interval $[F^{-1}(u), \infty)$ that contains its left endpoint, by monotonicity and right continuity; so $F(F^{-1}(u)) \geq u$. If $F^{-1}(u) \leq t$ then $u \leq F(F^{-1}(u)) \leq F(t)$. Conversely, if $u \leq F(t)$ then t belongs to the set, so $F^{-1}(u) \leq t$. Hence $P(F^{-1}(U) \leq t) = P(U \leq F(t)) = F(t)$.

(b) First, $P\{F(X) \leq F(t)\} = F(t)$ for every t: the event equals $\{X \leq t\} \cup \{X > t, F(X) = F(t)\}$, and the second part is a set where F is flat, which has probability 0. For $u \in (0,1)$, continuity gives t with $F(t) = u$, so $P\{F(X) \leq u\} = u$.

(c) By (a), we may take $X = F^{-1}(U)$, and $F(F^{-1}(U)) \geq U$. Hence $\{F(X) \leq y\} \subset \{U \leq y\}$. ■

Example 9.1

- Exponential, $F(x) = 1 - e^{-\lambda x}$: $F^{-1}(u) = -\frac{1}{\lambda}\log(1-u)$, and since $1 - U \overset{d}{=} U$ one uses $-\frac{1}{\lambda}\log U$.
- Cauchy(μ, β): $F(x) = \frac{1}{2} + \frac{1}{\pi}\arctan\frac{x-\mu}{\beta}$, $F^{-1}(u) = \mu + \beta\tan(\pi u - \frac{\pi}{2})$.
- Discrete uniform on $\{1, \dots, k\}$: $F^{-1}(u) = \lceil ku \rceil$.
- Geometric(p) on $\{1, 2, \dots\}$: if $Y \sim \text{Exp}(\lambda)$ with $1 - e^{-\lambda} = p$, then $\lceil Y \rceil$ is geometric, so $X = \lceil \log U / \log(1-p) \rceil$.

9.3 Normal random numbers

Φ^{-1} has no closed form (though accurate rational approximations exist).

Box–Muller. For independent $X, Y \sim N(0,1)$ in polar coordinates (R, Θ), the joint density is $\frac{1}{2\pi}e^{-r^2/2}r$, so $\Theta \sim \text{Unif}(0, 2\pi)$ independently of R, and $P(R^2 > s) = e^{-s/2}$, i.e. $R^2 \sim \text{Exp}$ with mean 2. Hence for independent $U, V \sim \text{Unif}(0,1)$,

$$R = \sqrt{-2\log U}, \quad \Theta = 2\pi V, \qquad (X, Y) = (R\cos\Theta, R\sin\Theta)$$

are independent standard normals.

Marsaglia's polar method. If (U, V) is uniform on the unit disk, written $(T\cos\Phi, T\sin\Phi)$, then $P(T^2 \le t, \Phi \le \phi) = \frac{\pi t \cdot \phi/(2\pi)}{\pi} = t\frac{\phi}{2\pi}$, so $T^2 \sim \text{Unif}(0,1)$ and $\Phi \sim \text{Unif}(0, 2\pi)$ independently. So $-2\log T^2$ plays the role of R^2, and $(U/T, V/T)$ that of $(\cos\Theta, \sin\Theta)$:

(1) draw U, V i.i.d. Unif$(-1, 1)$;
(2) if $T^2 = U^2 + V^2 \ge 1$, return to (1);
(3) output $X = U\sqrt{-2\log T^2/T^2}$ and $Y = V\sqrt{-2\log T^2/T^2}$.

It avoids trigonometric functions at the price of using $4/\pi$ times as many pairs (the rejection step is an instance of Section 9.5).

9.4 Generating from definitions

- Bernoulli(p): $\mathbf{1}\{U \le p\}$. Binomial(n, p): sum of n Bernoullis. Negative binomial(r, p): sum of r geometrics.
- Poisson(λ): the number N with $\prod_{i=1}^{N} U_i \ge e^{-\lambda} > \prod_{i=1}^{N+1} U_i$. Indeed $T_i = -\log U_i/\lambda$ are i.i.d. Exp(λ) interarrival times, and N is the number of arrivals of a rate-λ Poisson process in $[0, 1]$: $T_1 + \cdots + T_N \le 1 < T_1 + \cdots + T_{N+1}$.
- $\chi^2(\nu)$: $\sum_{i \le \nu} Z_i^2$; for even ν, also $-2\log\prod_{i \le \nu/2} U_i$, since $\chi^2(\nu) = \text{Gamma}(\nu/2, 2)$.
- $t(\nu) = Z/\sqrt{X/\nu}$ and $F(\nu_1, \nu_2) = (X_1/\nu_1)/(X_2/\nu_2)$ from independent normal and chi-square variates.

9.5 Acceptance–rejection sampling

To sample from a density f with no tractable cdf (von Neumann):

(1) find a density g that is easy to sample and a constant c with $f \le cg =: h$ (an *envelope*);
(2) draw $X \sim g$ and an independent $U \sim \text{Unif}(0,1)$;
(3) accept X if $U \le f(X)/h(X)$; otherwise return to (2).

The **body** of a nonnegative function f is $B_f = \{(x,y) : 0 \le y \le f(x)\}$, with area $|B_f| = \int f$.

Theorem 9.3

If $X \sim g$ and $U \sim \text{Unif}(0,1)$ are independent, then $(X, cg(X)U)$ is uniform on B_{cg}.

Proof. Given $X = x$, $Y = cg(x)U$ is uniform on $[0, cg(x)]$, with density $\frac{1}{cg(x)}\mathbf{1}\{0 \le y \le cg(x)\}$. Multiplying by $g(x)$, the joint density is $\frac{1}{c}\mathbf{1}_{B_{cg}}(x,y)$; and $|B_{cg}| = c$. ■

Corollary 9.4

An accepted X has density f, and each proposal is accepted with probability $1/c$.

Proof. Acceptance is the event $(X,Y) \in B_f \subset B_{cg}$. Conditioning a uniform distribution on a subset gives the uniform distribution on the subset, so the accepted (X,Y) is uniform on B_f, with density $\mathbf{1}_{B_f}/|B_f|$. Its x-marginal is $\int_0^{f(x)} dy/|B_f| = f(x)/\int f = f(x)$. The acceptance probability is $|B_f|/|B_{cg}| = 1/c$. ■

So the envelope should be tight (c close to 1); the number of proposals per accepted draw is geometric with mean c. f need only be known up to a constant, as long as the envelope uses the same constant.

Example 9.2 (Uniform on the disk)

In Marsaglia's method, $f = \frac{1}{\pi}\mathbf{1}\{u^2+v^2<1\}$ and $g = \frac{1}{4}\mathbf{1}_{[-1,1]^2}$, so $c = 4/\pi$ and $f/(cg)$ is the indicator of the disk: accept iff $U^2+V^2<1$.

Example 9.3 (Gamma variates)

Since $c\,\text{Gamma}(\alpha,1) = \text{Gamma}(\alpha,c)$ and $\text{Gamma}(\alpha,1) \overset{d}{=} \text{Gamma}(\lfloor\alpha\rfloor,1) \oplus \text{Gamma}(\alpha - \lfloor\alpha\rfloor,1)$, where the first is a sum of exponentials, it suffices to handle $0<\alpha<1$. Then $x^{\alpha-1}e^{-x} \le x^{\alpha-1}$ on $[0,1]$ and $\le e^{-x}$ on $(1,\infty)$, giving the envelope $h(x) = \{x^{\alpha-1}\mathbf{1}_{[0,1]} + e^{-x}\mathbf{1}_{(1,\infty)}\}/\Gamma(\alpha)$ with $c = (\frac{1}{\alpha}+\frac{1}{e})/\Gamma(\alpha)$. The proposal density $g = h/c$ has cdf $G(x) = x^\alpha/(1+\alpha/e)$ on $[0,1]$ and $1 - \alpha e^{-x}/(1+\alpha/e)$ for $x>1$, which inverts in closed form:

$$G^{-1}(u) = \begin{cases} \{(1+\alpha/e)u\}^{1/\alpha}, & u \le 1/(1+\alpha/e), \\ -\log\{(1/\alpha+1/e)(1-u)\}, & \text{otherwise.} \end{cases}$$

9.6 Multivariate distributions

Batch methods.

- Multinomial$(n; p_1,\dots,p_k)$: draw n categories, category j when $U \in [\sum_{l<j} p_l, \sum_{l\le j} p_l)$, and count. (Alternatively, independent Poissons with means λp_j conditioned on their sum being n, which is inefficient.)
- Multivariate normal $N(\mu,\Sigma)$: find L with $LL^\top = \Sigma$ and set $\mu + LZ$, $Z \sim N(0, I_p)$. Use the Cholesky factor (cheapest) or the symmetric square root $Q\Lambda^{1/2}Q^\top$ from the eigendecomposition (which also handles singular Σ). For large p the factorization is the bottleneck.
- Multivariate $t_\nu(\mu,\Sigma)$: $\mu + X/\sqrt{W/\nu}$ with $X \sim N(0,\Sigma)$ independent of $W \sim \chi^2(\nu)$.

Sequential (conditional) sampling. Draw components one at a time from their conditional distributions:

- Multinomial: $X_1 \sim B(n, p_1)$, then $X_2 \mid X_1 = x_1 \sim B(n - x_1, p_2/(1 - p_1))$, and so on.
- Multivariate normal: $X_1 \sim N(\mu_1, \sigma_{11})$, then $(X_2, \ldots, X_p) \mid X_1 = x_1 \sim N\big(\mu_2 + \Sigma_{21}(x_1 - \mu_1)/\sigma_{11},\ \Sigma_{22} - \Sigma_{21}\Sigma_{12}/\sigma_{11}\big)$, and recurse.